Lecture 5

Compatible and incompatible observables

Objectives. *In Lecture 1, a reading along x undid a reading along z. This lecture finds when two readings do not disturb each other. Four conditions turn out to be the same: the order of two readings does not matter, a second reading never changes the first, an unrecorded reading never changes the statistics of the other, and the two observables commute. Commuting observables can be read in a single measurement, and enough of them pick out one state, whatever the system was before. For observables that do not commute, the spreads of their readings in one state obey a lower bound set by their commutator.*

By the end of this lecture you will be able to:

- Find the probabilities of two readings made in either order, and of a reading repeated after another.
- Decide whether two observables are compatible from their commutator.
- Find the joint projections of commuting observables, and decide whether they form a complete set.
- Find the spreads of two observables in a state, and compare their product with the bound of the uncertainty relation.

5.1 Two readings in turn

Let $A = \sum_a a\,E_a$ and $B = \sum_b b\,G_b$ be two observables of an n-level system. We write G_b for the projections of B, since the letters P and Q are kept for the momentum and the position of a particle. Suppose that A is measured and then, at once, B, with no evolution between the two readings. By Theorem 3.4 with the branch operator $K = G_b E_a$, and by the cyclicity of the trace and $G_b^2 = G_b$, the readings "a, then b" occur with probability

$$\text{Prob}(a \text{ then } b) = \text{Tr}\big(G_b E_a \rho E_a G_b\big) = \text{Tr}\big(\rho\, E_a G_b E_a\big). \tag{5.1}$$

In the other order,

$$\text{Prob}(b \text{ then } a) = \text{Tr}\big(\rho\, G_b E_a G_b\big). \tag{5.2}$$

The third fact of Lecture 1 shows that the two can differ (Figure 1.2). Take a qubit in the state $|\uparrow\rangle$. Let $A = \sigma_z$ and $B = \sigma_x$. Measured first, σ_z reads +1 with certainty, and then σ_x reads +1 with probability $\frac{1}{2}$. So "+1 along z, then +1 along x" has probability $\frac{1}{2}$. Measured first, σ_x reads +1 with probability $\frac{1}{2}$ and leaves $|{+x}\rangle$, which then reads +1 along z with probability $\frac{1}{2}$. So "+1 along x, then +1 along z" has probability $\frac{1}{4}$. The order matters. It

matters because the reading along x replaces the state, and with it the value along z that the first magnet had made certain.

The physics

An ideal linear polariser passes light polarised along its axis. Light polarised at the angle α to the axis passes with the fraction $\cos^2\alpha$ of its intensity, which is Malus's law. The polarisation of a photon travelling in a fixed beam is a qubit. Transmission through a polariser at the angle α selects one reading of a polarisation observable, whose axis lies on a great circle of the Bloch sphere at the angle 2α. Indeed $\cos^2\alpha = \frac{1}{2}(1+\cos 2\alpha)$ is Born's rule (1.8) for two axes at the angle 2α. The absorbed photons are lost, not kept. Two crossed polarisers pass no light. Put a third one between them, at 45°. Half of an unpolarised beam passes the first polariser, half of that passes the middle one, and half again the last, so $\frac{1}{8}$ of the light comes through. The middle polariser reads an observable that is not compatible with the other two, as the magnet along x does in Figure 1.2. Sakurai and Napolitano's Fig. 1.4 shows the arrangement.

5.2 Compatibility is commutation

Definition 5.1 (Compatible observables)**.** Two observables A and B are *compatible* if, for every density matrix ρ and all readings a of A and b of B,

$$\text{Prob}(a \text{ then } b) = \text{Prob}(b \text{ then } a).$$

The definition asks for the equality in every state. § 5.5 shows that agreement in one state does not establish compatibility. The next theorem gives a test that uses no state at all.

Theorem 5.2 (Compatible observables commute)**.** *For two observables $A = \sum_a a\,E_a$ and $B = \sum_b b\,G_b$ on $\mathbb{C}^n$, the following are equivalent.*

1. $[A, B] = 0$.
2. *A and B are compatible.*
3. *A reading of B never changes the reading of A before it: for every density matrix, a measurement of A, then of B, then of A again, gives the same reading of A twice, with probability* 1.
4. *A reading of A that is not recorded does not change the statistics of B: for every density matrix ρ and every b, $\text{Tr}(\rho' G_b) = \text{Tr}(\rho G_b)$, where $\rho' = \sum_a E_a \rho E_a$ is the density matrix after the unrecorded reading* (3.8).

The readings are made at once, with no evolution between them.

Proof. We prove that the first condition gives the other three, and that each of the other three gives the first.

The first gives the others. Let $AB = BA$. By Lemma 4.3, applied to the self-adjoint matrix A, the matrix B commutes with every E_a. Applied again, to B and E_a, the lemma shows that every E_a commutes with every G_b. Then $E_a G_b E_a = G_b E_a = G_b E_a G_b$, and the probabilities (5.1) and (5.2) are equal. For the third condition, the readings a, b, a' have the branch operator $E_{a'} G_b E_a = G_b E_{a'} E_a$, which is 0 for $a' \neq a$. So the second reading of A is a with probability 1. For the fourth, by the cyclicity of the trace, $E_a G_b = G_b E_a$ and $\sum_a E_a = \mathbb{1}$,

$$\text{Tr}(\rho' G_b) = \text{Tr}\Big(\rho \sum_a E_a G_b E_a\Big) = \text{Tr}\Big(\rho\, G_b \sum_a E_a\Big) = \text{Tr}(\rho G_b).$$

The second gives the first. Fix a and b, and put $M = E_aG_bE_a - G_bE_aG_b$, a self-adjoint matrix. Applied to the pure states $\rho = |\psi\rangle\langle\psi|$, the second condition says $\langle\psi, M\psi\rangle = 0$ for every unit vector ψ. By the spectral theorem (*Linear Algebra*, Thm 10.10), each eigenvalue of M is such a value at a unit eigenvector, so every eigenvalue is 0 and $M = 0$. Now let v be a vector with $E_av = 0$. Then, since $M = 0$,

$$\|E_aG_bv\|^2 = \langle v, G_bE_aG_bv\rangle = \langle v, E_aG_bE_av\rangle = 0.$$

Every vector $(\mathbb{1} - E_a)w$ is such a v, so $E_aG_b(\mathbb{1} - E_a) = 0$, that is, $E_aG_b = E_aG_bE_a$. Taking adjoints gives $G_bE_a = E_aG_bE_a$. Hence $E_aG_b = G_bE_a$.

The third gives the first. For $a' \neq a$ and every unit vector ψ, the branch a, b, a' has probability $\|E_{a'}G_bE_a\psi\|^2 = 0$ in the state $[\psi]$, so $E_{a'}G_bE_a = 0$. Summing over $a' \neq a$ gives $(\mathbb{1} - E_a)G_bE_a = 0$, that is, $G_bE_a = E_aG_bE_a$. Taking adjoints gives $E_aG_b = E_aG_bE_a$, and again $E_aG_b = G_bE_a$.

The fourth gives the first. By the cyclicity of the trace, the fourth condition says $\mathrm{Tr}\big(\rho\,(\sum_a E_aG_bE_a - G_b)\big) = 0$ for every ρ. As for the second, this makes the self-adjoint matrix $\sum_a E_aG_bE_a - G_b$ zero. Multiplying $\sum_a E_aG_bE_a = G_b$ by E_c on the left gives $E_cG_bE_c = E_cG_b$, and on the right $E_cG_bE_c = G_bE_c$. So $E_cG_b = G_bE_c$ for every c.

In every case every E_a commutes with every G_b. Hence $A = \sum_a a\,E_a$ commutes with $B = \sum_b b\,G_b$. ∎

So compatibility, a statement about every experiment, is decided by one commutator of two matrices. The spin readings σ_z and σ_x of § 5.1 are not compatible, since $\sigma_z\sigma_x - \sigma_x\sigma_z = 2i\sigma_y \neq 0$. On a qubit, by the Pauli rule, two observables $a\mathbb{1} + b\,\mathbf{m}\cdot\boldsymbol{\sigma}$ and $a'\mathbb{1} + b'\,\mathbf{m}'\cdot\boldsymbol{\sigma}$ with $b > 0$ and $b' > 0$ commute exactly when $\mathbf{m}' = \pm\mathbf{m}$. A qubit has no compatible readings along two different axes. The fourth condition shows it once more. Keep both beams that leave a magnet along x, and record neither reading. The atoms that read +1 along z before that magnet then read −1 along z with probability $\frac{1}{2}$, by (3.8).

5.3 Reading two observables at once

Compatible observables can be read together, by one measurement.

Proposition 5.3 (A joint reading)**.** *Let $A = \sum_a a\,E_a$ and $B = \sum_b b\,G_b$ commute. The matrices E_aG_b are orthogonal projections with $\sum_{a,b} E_aG_b = \mathbb{1}$. A measurement of A followed at once by one of B gives the readings a, b with probability $\mathrm{Tr}(\rho\,E_aG_b)$ and, if this is positive, leaves*

$$\frac{E_aG_b\,\rho\,E_aG_b}{\mathrm{Tr}(\rho\,E_aG_b)}.$$

These are the probabilities and the states of a single ideal measurement whose projections are the nonzero E_aG_b.

Proof. By the proof of Theorem 5.2, $E_aG_b = G_bE_a$. So $(E_aG_b)^\dagger = G_bE_a = E_aG_b$ and $(E_aG_b)^2 = E_a^2G_b^2 = E_aG_b$, and $\sum_{a,b} E_aG_b = \big(\sum_a E_a\big)\big(\sum_b G_b\big) = \mathbb{1}$. The branch operator of the sequence is $G_bE_a = E_aG_b$, and Theorem 3.4 gives the rest. ∎

For example, a measurement of an observable $\sum_{a,b} c_{ab}\,E_aG_b$ with distinct real numbers c_{ab} is a single measurement with these projections. The range of E_aG_b consists of the vectors v with $Av = av$ and $Bv = bv$. *Linear Algebra*, Lecture 10, shows that commuting self-adjoint matrices have a common orthonormal eigenbasis, which is the same fact in another form.

When every joint projection has rank 1, a joint reading fixes the state.

Definition 5.4 (Complete set). Commuting observables $A_1, \dots, A_r$ form a *complete set* if every nonzero product $E^{(1)}_{a_1} \cdots E^{(r)}_{a_r}$ of their projections has rank 1.

By Proposition 5.3, applied in turn, the products are projections. If the readings $a_1, \dots, a_r$ have positive probability, their projection is $|\chi\rangle\langle\chi|$ for a unit vector χ. By the rule of § 3.4, and since $|\chi\rangle\langle\chi|\,\rho\,|\chi\rangle\langle\chi| = \langle\chi, \rho\chi\rangle\,|\chi\rangle\langle\chi|$ and $\operatorname{Tr}(\rho\,|\chi\rangle\langle\chi|) = \langle\chi, \rho\chi\rangle$, the state afterwards is

$$\frac{|\chi\rangle\langle\chi|\,\rho\,|\chi\rangle\langle\chi|}{\operatorname{Tr}(\rho\,|\chi\rangle\langle\chi|)} = |\chi\rangle\langle\chi|\,.$$

So a joint reading of a complete set prepares the ray $[\chi]$, whatever the state was before. The readings of a complete set label the states of an orthonormal basis. Labels of this kind are called *quantum numbers*. A single observable with n distinct readings is a complete set by itself. *Linear Algebra*, Ex. 10.39, previews the definition.

5.4 Three readings of a three-level system

Example 5.5 (The energy, Z and X). Take the three-level system of § 4.1, with the observables $Z = \operatorname{diag}(1, 0, -1)$ and $X = |e_+\rangle\langle e_-| + |e_-\rangle\langle e_+|$ of § 4.6.

1. *Who commutes with whom.* Both Z^2 and X^2 equal $|e_+\rangle\langle e_+| + |e_-\rangle\langle e_-|$, so the Hamiltonian (4.1) is $H = \varepsilon Z^2 = \varepsilon X^2$. A function of an observable commutes with it, so H commutes with Z and with X. But

$$ZX - XZ = 2\big(|e_+\rangle\langle e_-| - |e_-\rangle\langle e_+|\big) \neq 0,$$

   so Z and X are not compatible.
2. *Two ways to refine the energy.* Each of Z and X has three distinct readings, so each is a complete set by itself. A reading of Z splits the degenerate level ε into the rays e_+ and e_-, and a reading of X splits it into χ_+ and χ_-. Both are compatible with the energy, and they are not compatible with each other.
3. *Order matters.* Prepare e_+. A reading of Z, then of X, gives $(+1, +1)$ with probability $1 \cdot \frac{1}{2} = \frac{1}{2}$, by (5.1). A reading of X, then of Z, gives $X = +1$ with probability $|\langle\chi_+, e_+\rangle|^2 = \frac{1}{2}$ and leaves χ_+, which gives $Z = +1$ with probability $\frac{1}{2}$. In total, the probability is $\frac{1}{4}$.
4. *A reading undone.* Prepare e_+, and measure Z, then X, then Z. The first reading of Z is $+1$. After either reading ± 1 of X, the state is $\chi_\pm$, which gives $Z = -1$ with probability $\frac{1}{2}$. So the second reading of Z differs from the first with probability $\frac{1}{2}$. This is the third fact of Lecture 1, on a three-level system.
5. *A shared eigenvector.* Both Z and X read 0 on e_0. A system prepared in e_0 gives the same joint probabilities in either order, and keeps its reading of Z through a reading of X. Yet Z and X are not compatible, since the state e_+ above shows that the order matters.

5.5 What commutation does not say

The shared eigenvector of Example 5.5 is a warning. A common eigenvector does not make two observables compatible. Compatibility asks for the equality of Definition 5.1 in every state, and e_0 is one state. In the same way, a state may give certain readings of

two incompatible observables, as e_0 does for Z and X. Compatibility is a property of two observables, not of an observable and a state.

For a qubit, the readings along two axes obey an exact bound. By Born's rule on the ball (§ 3.4), a reading along $\mathbf{m}$ gives ± 1 with probability $\frac{1}{2}(1 \pm \mathbf{m} \cdot \mathbf{r})$, so the difference of the two probabilities is $\mathbf{m} \cdot \mathbf{r}$.

Proposition 5.6 (Readings along z and x)**.** *For every density matrix of a qubit, with Bloch vector* $\mathbf{r}$*,*

$$\begin{aligned}\big(\text{Prob}(+1 \textit{ along } z) - \text{Prob}(-1 \textit{ along } z)\big)^2 + \big(\text{Prob}(+1 \textit{ along } x) - \text{Prob}(-1 \textit{ along } x)\big)^2 \\ = r_z^2 + r_x^2 \le 1,\end{aligned}$$

with equality if and only if the state is pure and $r_y = 0$*.*

Proof. The two differences are r_z and r_x, and $r_x^2 + r_z^2 \le |\mathbf{r}|^2 \le 1$ by Theorem 3.6. Equality holds exactly when $r_y = 0$ and $|\mathbf{r}| = 1$, that is, for a pure state in the xz-plane. ∎

So a state whose reading along z is certain, $r_z = \pm 1$, gives ± 1 along x with probability $\frac{1}{2}$ each.

Theorem 5.2 is about readings made at once. If the system evolves between them, a reading of A at one time and of A again later can disagree even with no reading in between. Whether a reading survives the evolution is the question of Lecture 6.

For any two observables, § 5.6 bounds the spreads of their readings in one state.

5.6 The spreads of two readings

An ideal measurement of A on many systems prepared in the state ρ gives readings that scatter about their mean $\langle A \rangle = \text{Tr}(\rho A)$. Their *spread* $\Delta A \ge 0$ is given by

$$(\Delta A)^2 = \sum_a \big(a - \langle A \rangle\big)^2 \,\text{Tr}(\rho E_a) = \text{Tr}(\rho A^2) - \langle A \rangle^2. \tag{5.3}$$

The second form follows by expanding the square, since $\sum_a a^2 E_a = A^2$ and $\sum_a \text{Tr}(\rho E_a) = 1$. The spread is 0 exactly when one reading is certain. *Linear Algebra* writes ΔA for the matrix $A - \langle A \rangle \mathbb{1}$. Here ΔA is a number.

Theorem 5.7 (The uncertainty relation)**.** *For observables* A *and* B *and every density matrix* ρ*,*

$$\Delta A \, \Delta B \ge \frac{1}{2} \left| \text{Tr}(\rho \, [A, B]) \right| .$$

Proof. For a pure state this is *Linear Algebra*, Prop. 9.36. In general, write $\rho = \sum_k p_k \, |u_k\rangle\langle u_k|$ with unit vectors u_k and weights $p_k \ge 0$, for instance by the spectral theorem. Put $A_0 = A - \langle A \rangle \mathbb{1}$ and $B_0 = B - \langle B \rangle \mathbb{1}$. They are self-adjoint, and $[A_0, B_0] = [A, B]$. Consider the number

$$z = \text{Tr}(\rho A_0 B_0) = \sum_k p_k \, \langle A_0 u_k, B_0 u_k \rangle \, .$$

Since $\overline{\langle A_0 u_k, B_0 u_k \rangle} = \langle B_0 u_k, A_0 u_k \rangle = \langle u_k, B_0 A_0 u_k \rangle$, the conjugate $\bar{z}$ is $\text{Tr}(\rho B_0 A_0)$. Hence $\text{Tr}(\rho \, [A, B]) = z - \bar{z}$, and $|\text{Tr}(\rho \, [A, B])| \le 2 \, |z|$. By the Cauchy–Schwarz inequality, first for

each pair of vectors and then for the sum over k,

$$|z| \le \sum_k p_k \, \|A_0 u_k\| \, \|B_0 u_k\| \le \Big(\sum_k p_k \, \|A_0 u_k\|^2\Big)^{1/2} \Big(\sum_k p_k \, \|B_0 u_k\|^2\Big)^{1/2} = \Delta A \, \Delta B,$$

since $\sum_k p_k \, \|A_0 u_k\|^2 = \mathrm{Tr}(\rho A_0^2) = (\Delta A)^2$, and likewise for B. ∎

The spreads belong to separate runs. On some of the systems A is read, and on others B. The theorem says nothing about a joint reading of A and B, which does not exist when they do not commute, nor about one reading disturbing the other, which is the third condition of Theorem 5.2.

For a qubit, take $A = \sigma_x$ and $B = \sigma_y$. Since $\sigma_x^2 = \sigma_y^2 = \mathbb{1}$ and $\big[\sigma_x, \sigma_y\big] = 2\mathrm{i}\sigma_z$, we have $(\Delta\sigma_x)^2 = 1 - r_x^2$, $(\Delta\sigma_y)^2 = 1 - r_y^2$ and $\mathrm{Tr}(\rho \, \big[\sigma_x, \sigma_y\big]) = 2\mathrm{i} r_z$. So the theorem reads

$$(1 - r_x^2)(1 - r_y^2) \ge r_z^2.$$

The difference of the two sides is $(1 - |\mathbf{r}|^2) + r_x^2 r_y^2$, so equality holds exactly for the pure states with $r_x r_y = 0$. The bound depends on the state, and it can vanish for observables that do not commute. For example, the state $|{+x}\rangle$ has $\Delta\sigma_x = 0$ and $r_z = 0$. On the three-level system, the state e_0 has $\Delta Z = \Delta X = 0$. The state $\psi = \frac{1}{\sqrt{2}}(e_+ + \mathrm{i}e_-)$ has $\Delta Z = \Delta X = 1$ and $\mathrm{Tr}(|\psi\rangle\langle\psi| \, [Z, X]) = 2\mathrm{i}$, so it reaches the bound. Proposition 5.6 is a different statement. It limits $r_z^2 + r_x^2$ also in the states where $\mathrm{Tr}(\rho \, [\sigma_z, \sigma_x]) = 2\mathrm{i} r_y$ is 0.

Two observables of an n-level system

Compatible: A and B give the same probabilities in either order, for every state, if and only if $[A, B] = 0$. Then a reading of B never changes the reading of A before it, and an unrecorded reading of A leaves the statistics of B alone (Theorem 5.2). *On a qubit:* $r_z^2 + r_x^2 \le 1$ (Proposition 5.6). *Read together:* commuting observables are read by one measurement with the projections $E_a G_b$ (Proposition 5.3). *Complete set:* commuting observables whose nonzero joint projections have rank 1. Their joint reading leaves a ray fixed by the readings alone. *Spreads:* $\Delta A \, \Delta B \ge \frac{1}{2} \, |\mathrm{Tr}(\rho \, [A, B])|$, for separate runs of one preparation (Theorem 5.7).

Summary.

- *Compatible observables commute*: the order of two readings never matters, a second reading never changes the first, and an unrecorded reading never changes the statistics of the other, exactly when the two observables commute (Theorem 5.2).
- *Readings along z and x*: on a qubit, $r_z^2 + r_x^2 \le 1$ (Proposition 5.6).
- *A joint reading*: commuting observables are read by a single measurement, and a complete set of them leaves one ray (Proposition 5.3, Definition 5.4).
- *The uncertainty relation*: in every state, $\Delta A \, \Delta B \ge \frac{1}{2} \, |\mathrm{Tr}(\rho \, [A, B])|$ (Theorem 5.7).

Lecture 6 lets the system evolve between the readings, and finds that the observables compatible with the Hamiltonian are the conserved ones.

References

Sakurai and Napolitano, *Modern Quantum Mechanics*, §1.4.3–1.4.4, pp. 27–31, and §1.1.3, pp. 6–7. Cohen-Tannoudji, Diu and Laloë, *Quantum Mechanics*, vol. 1, Ch. II, §D-3-b,

pp. 137–139, and Ch. III, §C-6, pp. 232–235. Sakurai and Napolitano, §1.4.5, pp. 31–33 (the uncertainty relation). Nielsen and Chuang, *Quantum Computation and Quantum Information*, Box 2.4, p. 89.